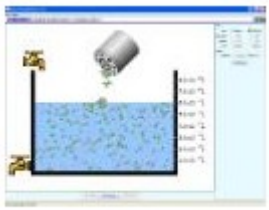
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Introduction to Solubility PhET Lab (created 3/2011)

We can express concentrations in terms of moles solute per liters of solution

Introduction


Salts & Solubility

When we cook pasta, we frequently dissolve table salt in water. Is there a limit to how much salt we can get into the water?

When ionic-bonded compounds (like table salt, NaCl) are dissolved in polar solvents (like water), the **polar** water molecules act to pull the ions in the compound apart. When this happens (called dissociation), ions are released into the solution. Is there a limit to the number of ions that can exist in solution?

Pre-lab Vocabulary

Please use your textbook, your notes, or the internet to define the following terms:

Saturated _____

Unsaturated _____

Supersaturated _____

Molarity _____

Cation _____

Anion _____

Dissociation _____

Solubility _____

Procedure *PhET Simulations → Play With Sims → Chemistry → Salts and Solubility →* Run Now!

Table Salt Take some time and familiarize yourself with the simulation before answering the below questions:

1. What happens to the bound Na+ and Cl- ions as they enter the water?

   _____

2. Is there a point at which no more sodium and chloride ions will dissolve? How do you know?

   _____

3. What do we call a solution that has as many ions as it can hold?

   _____

4. What happens if you add additional, solid NaCl after the maximum has been reached?

   _____

5. What happens when more water is added to a *saturated* solution?

   _____

6. What happens to the ions bound together as a solid when the solution's volume is reduced?

   _____

7. How many total Na^+ ions can you add until no more will dissolve (the maximum amount)?

   _____

8. How many **moles** does this many ions represent? (recall, 1 mole = 6.02×10^{23} things)

   _____

9. What **volume** (in **Liters**) did you use to get the saturation point in #7 above?

   _____

10. What **concentration** is this? (In **mole/L**, use #8 and #9's answers) **This is molarity, M.**

    _____





Slightly Soluble Salts

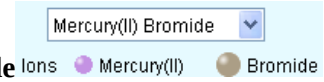
For each slightly soluble salt…

1. Determine the Chemical Formula
2. Write a Dissociation Equation (Example: $NaCl_{(s)} \rightarrow Na^{+}_{(aq)} + Cl^{-}_{(aq)}$)
3. Determine Concentration (mole/L) of the molecule and EACH ION in the compound in a saturated solution.

Remember…to find concentration in Molarity: calculate moles present, divide moles into volume in Liters.

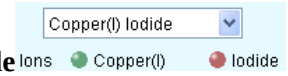
↓Scratch area to show your work↓

Mercury (II) Bromide



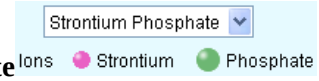
1. Chemical Formula: _____
2. Dissociation Equation: _____
3. Concentration of entire molecule at saturation (M): _____
4. Concentration of cation (+) at saturation (M): _____
5. Concentration of anion (-) at saturation (M): _____

Copper (I) Iodide



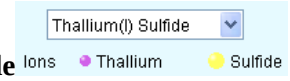
6. Chemical Formula: _____
7. Dissociation Equation: _____
8. Concentration of entire molecule at saturation (M): _____
9. Concentration of cation (+) at saturation (M): _____
10. Concentration of anion (-) at saturation (M): _____

Strontium Phosphate



11. Chemical Formula: _____
12. Dissociation Equation: _____
13. Concentration of entire molecule at saturation (M): _____
14. Concentration of cation (+) at saturation (M): _____
15. Concentration of anion (-) at saturation (M): _____

Thallium Sulfide



16. Chemical Formula: _____
17. Dissociation Equation: _____
18. Concentration of entire molecule at saturation (M): _____
19. Concentration of cation (+) at saturation (M): _____
20. Concentration of anion (-) at saturation (M): _____

Name:_____

Post-Lab Conclusion Questions and Calculations

1. Adding water **to a solution** of ions *increases / decreases / doesn't change* concentration.
2. Reducing volume **of an entire solution** *increases / decreases / doesn't change* concentration.
3. Adding more solid compound to an **unsaturated** solution *increases / decreases / doesn't change* concentration.
4. Adding more solid compound to a **saturated** solution *increases / decreases / doesn't change* concentration.
5. The bound ions of a solid compound at the bottom of a saturated solution *stay bound / dissolve and are rebound*. (watch carefully)
6. Determine the concentration of a solution of 2.4 moles of sugar (no dissociation) in 3.5 L of water

SHOW WORK

_____ M

7. How many moles of NaCl would be required to produce **.**95 L of a **.**58 M NaCl solution?

SHOW WORK

_____ mol

8. What volume of water would be required to dissolve 67 grams of Lithium Fluoride (LiF) to a concentration of 2.6 M?

SHOW WORK

_____ L

9. What is the concentration of a solution of 3.5 x10 ^{22} molecules of sugar in 25 mL of water?

SHOW WORK

_____ M

10. How many moles of $FeCl_3$ (compound) are present in 2.1 L of a **.**85 M solution?

SHOW WORK

_____ mol $FeCl_3$

11. How many moles of Fe^{3+} ions are present in this solution?

_____ mol Fe^{3+}

12. How many moles of Cl- are present in this solution?

_____mol Cl^-

13. What is the concentration of a solution made from 130 grams of $Cu_3(PO_4)_2$ in 3.9 L of water?

SHOW WORK

_____ M

14. What is the concentration of Copper (II) ions in the above solution?

_____ M

15. What is the concentration of Phosphate ions in the above solution?

_____ M